

SDM234-Homework1

Your name:

Your ID:

1. Let

$$z = e^{1+i\frac{2\pi}{3}} + e^{1-i\frac{\pi}{6}}.$$

Write the **real part**, **imaginary part**, **modulus**, and **argument** of z .

2. Find and graph all roots in the complex plane:

(1)

$$\sqrt[6]{-64}$$

(2)

$$\sqrt[4]{81}$$

3. **Parallelogram Equality.** Prove the following identity and explain why it is called the *parallelogram equality*:

$$|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2(|z_1|^2 + |z_2|^2).$$

4. Prove that

$$\sin 2\theta + \sin 4\theta + \cdots + \sin 2n\theta = \frac{\sin(n+1)\theta \sin n\theta}{\sin \theta}.$$

$$\cos 2\theta + \cos 4\theta + \cdots + \cos 2n\theta = \frac{\cos(n+1)\theta \sin n\theta}{\sin \theta}.$$

Hint for Problems 4: You may use the following relations:

$$\sin n\theta = \frac{e^{in\theta} - e^{-in\theta}}{2i}, \quad \cos n\theta = \frac{e^{in\theta} + e^{-in\theta}}{2},$$

and

$$\sum_{k=1}^n e^{i2k\theta} = \frac{(1 - e^{i2n\theta})e^{i2\theta}}{1 - e^{i2\theta}}.$$

Other methods are also encouraged.